PROJECT SYNOPSIS

Course: Database Management Systems

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Project Title: Campus Resource Booking & Waitlist Management System

1. Problem Statement

Educational institutions often manage shared resources such as seminar halls, laboratories, study rooms, and equipment through manual processes or disconnected systems. This can cause overlapping reservations, inefficient resource utilization, limited availability tracking, and difficulties handling cancellations. A centralized system is needed to streamline allocation, prevent booking conflicts, and ensure fair access to campus facilities.

2. Proposed Solution

The proposed web-based platform will allow students and authorized staff to view resource availability, make reservations, and manage bookings. It will detect scheduling conflicts and maintain a waitlist for fully booked resources. When a reservation is cancelled, eligible waitlisted users can be allocated the available slot according to predefined priority rules. An administrative dashboard will display resource utilization, booking trends, cancellations, and availability.

3. Key Features

- **Resource Booking:** Search and reserve facilities by date, time, and type.
- **Conflict Detection:** Prevent overlapping reservations through transactional validation and database constraints.
- **Automated Waitlist:** Maintain a priority queue and process cancellations automatically.
- **Role-Based Access:** Provide separate student and administrator functions with authentication.
- **Analytics & Audit:** Report peak hours, utilization, cancellations, booking history, and status changes.

4. Technology Stack

Frontend: Next.js **Backend:** Next.js API Routes / Route Handlers
Relational Database: MySQL **NoSQL Database:** MongoDB **Version Control:** Git and GitHub

5. Database Design & Advanced Concepts

MySQL will store structured information such as users, resources, reservations, and waitlist entries, using primary keys, foreign keys, relationships, and integrity constraints. MongoDB will store flexible booking activity logs, cancellation records, and administrative audit events.

- **Transactions:** Keep booking and waitlist allocation operations atomic.
- **Triggers:** Record relevant booking status changes and enforce database-side rules.
- **Stored Procedures:** Handle reservation validation and waitlist processing.
- **Views & Indexing:** Support availability searches and utilization reports.
- **MongoDB Aggregation:** Analyze booking activity and audit records.

6. Expected Outcome

The system aims to provide a centralized, reliable platform for campus resource management. It will reduce booking conflicts, improve resource utilization, automate waitlist allocation, and provide useful administrative insights while demonstrating the integration of relational and NoSQL databases in a functional web application.